

AN ϵ -CONSTRAINT METHOD WITH ADAPTIVE MODE ELIMINATION SCHEME FOR MULTI-OBJECTIVE MIXED INTEGER PROGRAMMING

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Abstract

The significance of time-cost trade-off is well recognized for over five decades ever since the development of the critical path method. This trade-off originates from the fact that faster accomplishment of project activities, and thereby the project as a whole, is possible should there be more productive resources assigned. Increasing productivity, on the other hand, typically comes at a cost but possibly helps avoid incurrence of additional overhead and/or penalty expenses. The precise determination of the balance between the planned productivity and the aforesaid costs often poses a tough decision to make by the management. The process of finding the proper combination of execution modes gets particularly complex and computationally expensive for real-scale projects. Accordingly, to help set the proper balance between the time and cost of large projects, an ϵ -constraint method is presented in this study featuring a clever adaptive mode elimination scheme. The proposed approach involves the iterative optimal solution of the trade-off problem over the feasible project durations. For each project realization, the complexity of the optimization problem is first reduced with the aid of the adaptive mode elimination by discarding the redundant execution mode(s). This study uses mixed integer linear programming for formulation and development of the proposed model and uses a project example borrowed from the literature to experiment its usefulness.

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Keywords: ϵ -constraint method, mixed integer programming, mode elimination, multiobjective optimization, network reduction.

1. Introduction

The main objectives of most projects are completing them on time, within budget, and with high stakeholder satisfaction. In the construction industry, competition among companies often involves offering unique scheduling proposals that result in reasonable profits. Project stakeholders typically aim to complete projects as efficiently as possible to free up resources and gain financial benefits. There are often multiple alternatives available for execution of each project activity in terms of both duration and cost. Many activities can be performed using more efficient construction methods or by allocating additional resources for speeding them up. Decision-makers seek to expedite the project while minimizing extra expenses by finding the best range of alternatives for performing project activities and taking advantage of slack times in the project network. To achieve this, decision-makers strive to find the optimal balance between direct and indirect project costs since on the one hand, reduced project duration leads to a decrease in the project cost, on the other hand, the additional direct costs required for acceleration will outweigh the amount of indirect cost savings due to the duration reduction.

Achieving a suitable trade-off between the cost and duration of a project is thus an intricate and computationally challenging problem solving which necessitates taking a systematic approach. The issue of balancing time and cost in a project, which is commonly known as the time-cost trade-off problem (TCTP), has been widely recognized in the literature and studied under various assumptions. Within the body of literature, there are three common types of the time-cost trade-off problem (TCTP) that have been studied: the deadline problem, the budget problem, and the Pareto front problem. The goal of the deadline problem is to identify a set of time-cost alternatives that will minimize the total cost

of a project, including bonuses and penalties, while adhering to a given deadline [1,2]. The budget problem, on the other hand, seeks to minimize the duration of the project without exceeding the allotted budget [3]-[6]. Finally, the Pareto front problem, being the focus of this study, involves identifying the non-dominated time-cost profile over the set of feasible project durations to generate the efficient frontier known as the Pareto front [7].

The research area of TCTP first emerged in the 1960s, coinciding with the introduction of project network analysis techniques by Kelley and Walker [8], Fulkerson [9], and Kelley [10]. Since then, there has been significant progress in this domain, fueled by advancements in computer science and technology. There exists a substantial body of research that focuses on different variants of this problem and seeks to develop effective solutions that can help decision-makers navigate the complexities involved in balancing time and cost objectives. As such, it has attracted significant attention from scholars, researchers, and practitioners alike, who pursue developing practical solutions that can be applied in real-world situations. The ongoing efforts to develop more effective and efficient methods for managing the TCTP reflect the importance of this problem in project management and the significant impact it can have on a project's success. These methods mainly include heuristics, exact approaches, among others. Such methods differ in terms of their computational complexity, solution quality, and applicability to real-world situations. Due to their capability in locating the optima, many exact approaches have been used to model TCTP, such as linear programming proposed by Kelley [10], dynamic programming by Butcher [3], hybrid linear/integer programming by Liu et al. [11], and branch-and-bound algorithm by Demeulemeester et al. [2]. Though, it is a well-known fact that exact methods tend to require great computational effort. Accordingly, several researchers have attempted to simplify TCTPs through various reduction techniques. As such, the notion of reducing the size of TCTP through mode elimination is not a new concept and has been previously explored in the literature for both the deadline and the budget problems. The most noteworthy of the studies on problem size reduction are due to Akkan et al. [12], Hafizoglu and Azizoglu [13], who explored utility of mode elimination in the deadline TCTP, as well as Degirmenci and Azizoglu [14], who examined the use of a similar technique in the budget TCTP.

However, to the best of author's knowledge, this study is the first to examine the potential benefits of integrating mode elimination into the truly multi-objective Pareto front TCT problem. The proposed methodology employs a dynamic problem size reduction mechanism, which leads to the concept of adaptive mode elimination, and provides a novel contribution to the literature on TCTP research. By leveraging this approach, this study aims to enhance the efficiency and effectiveness of the solution procedure for Pareto front TCTPs, which are inherently complex and require decision-makers to consider multiple objectives instantaneously. Therefore, this study presents an ϵ -constraint method that incorporates an adaptive mode elimination technique to assist in achieving the optimal balance between project duration and cost specifically for large Pareto front TCTPs. The proposed approach entails an iterative process of solving the deadline TCTP over the complete set of feasible project durations via adjusting the ϵ value. At each level of ϵ , the size of the search space gets reduced with the aid of the clever adaptive mode elimination by detecting and discarding the redundant execution mode(s). The redundant modes are detected as the ones which cannot contribute to capturing of the current ϵ . The mixed integer linear programming (MILP) method is utilized to formulate and develop the proposed optimization model. In order to explore effectiveness of the present method, a benchmark TCTP is borrowed from the literature. Comparisons are made between the reduced and unreduced search spaces highlighting the usefulness and efficacy of the proposed optimization model.

The remainder of this paper is organized as follows. Section 2 elaborates on the steps of the proposed optimization model. In Section 3, the proposed approach is experimented for reduction of multi-objective search space. Finally, Section 4 provides conclusions.

2. Methodology

As an activity gets expedited, a decline in productivity is expected to happen after a certain point due to several reasons including congestion of the construction site. Accordingly, individual activities forming

a TCTP will lead to a convex solution space. Convexity of the solution space, on the other hand, facilitates converging to global optima even for large TCTPs using mathematical models, should there be no limitation on the processing computer's memory. In this study, the mathematical model is formulated using mixed integer linear programming (MILP) and Gurobi Solver version 9.5.1 is employed to solve the model. The proposed optimization procedure incorporates a particular strategy to reduce the problem size that helps solve complicated Pareto front problems with substantially reduced computational effort and time.

Mathematical formulation

The formulation used to construct MILP for Pareto front time-cost trade-off problem is given in Eqs. (1)-(7).

$$\text{minimize } \mathbf{f} = (f^{(1)}, f^{(2)}) \quad (1)$$

$$f^{(2)} = \sum_{j=1}^S \sum_{k=1}^{m(j)} (dc_{jk} x_{jk}) + f^{(1)} \times ic \quad (2)$$

subject to:

$$\sum_{k=1}^{m(j)} x_{jk} = 1, \quad \forall j \in \{1, \dots, S\} \quad (3)$$

$$\sum_{k=1}^{m(j)} d_{jk} x_{jk} + St_j \leq St_l, \quad \forall l \in Sc_j \quad \text{and} \quad \forall j \in \{1, \dots, S\} \quad (4)$$

$$f^{(1)} \leq \varepsilon \quad (5)$$

$$f^{(1)} \geq Ft_S \quad (6)$$

$$St_0 = 0 \quad (7)$$

where; $f^{(1)}$ denotes the first objective function, i.e., duration; $f^{(2)}$ denotes the second objective function, i.e., cost; dc_{jk} is the direct cost of k -th mode of j -th activity; x_{jk} is a binary variable which is set to 1 when k -th mode is selected for j -th activity, is set to 0 otherwise; ic denotes the rate of daily indirect cost; d_{jk} is the duration of k -th mode of j -th activity; St_j is the start time of j -th activity; Ft_j denotes the finish time of j -th activity; and Sc_j is the set of immediate successors for j -th activity. Eq. (1) defines the objective function for the Pareto front TCTP. Objective function of cost is calculated per Eq. (2). Constraint defined as Eq. (3) secures selection of only a single time-cost alternative at a time for each activity. The precedence relationships are sustained using Eq. (4). Constraints defined as Eqs. (5) and (6) represent the bounds on project duration based on ε 's value and the finish time of the last activity. Start time of the first activity is set using Eq. (7).

Adaptive mode elimination technique

As discussed earlier, reducing the dimensionality of a problem would exponentially decrease the search space, leading to a significant reduction in computational effort. One way to decrease the dimension of a TCTP is to systematically eliminate the execution modes that cannot contribute to achievement of a particular project duration. This problem size reduction technique is hailed as mode elimination which is established based on the preprocessing techniques described by Akkan et al. [12]. Mode elimination has been the subject of research studies focusing on deadline TCTP [12,13] and budget TCTP [14] but has never been deployed for solution of Pareto front TCTP. To address this gap in the literature, a clever adaptive mode elimination strategy is proposed in this study which specifically aids unravelling Pareto front TCTPs. The optimization model of the present study incorporates the adaptive mode elimination into the well-established ε -constraint method [15]. The ε -constraint-based method involves iterative solution of the deadline TCTP by sweeping ε from minimum possible duration to maximum possible duration in steps of one. At each level of ε , the adaptive mode elimination reduces the problem size by

discarding the redundant time-cost mode(s) as follows. Initially, for each activity, the associated all-crashed mode is selected and CPM calculations are carried out. Next, the so-called long modes are identified per Eqs. (8)-(10) and eliminated since they cannot contribute to capturing of the current ε . Discarding the so-called short modes at the current ε level ensues subsequent to elimination of the long modes as follows. For each activity, the corresponding all-normal mode is selected and CPM calculations are performed. Next, short modes are eliminated per Eqs. (11)-(13) since their contribution to achieving the current value of ε is null.

eliminating long modes:

$$\delta_C = \varepsilon - LF_S \quad (8)$$

$$LF_j = LF_j + \delta_C \quad (9)$$

$$Decision \begin{cases} \text{if } LF_j - ES_j + 2 \leq d_{jk} & \text{remove modes 1 to } k \\ \text{otherwise,} & \text{dismiss} \end{cases} \quad (10)$$

eliminating short modes:

$$\delta_N = \varepsilon - LF_S \quad (11)$$

$$LF_j = LF_j - \delta_N \quad (12)$$

$$Decision \begin{cases} \text{if } LF_j - ES_j + 1 > d_{jk} & \text{remove modes } k+1 \text{ to } m(j) \\ \text{otherwise,} & \text{dismiss} \end{cases} \quad (13)$$

$$\forall j \in \{1, \dots, S\} \quad \text{and} \quad \forall k \in \{1, \dots, m(j)\} \quad (14)$$

where; δ_C denotes the difference between ε and late finish time of the last activity in all-crashed schedule; LF_j represents the late finish time of j -th activity; ES_j is the early start time of j -th activity; d_{jk} denotes duration of k -th mode of j -th activity; and δ_N denotes the difference between ε and late finish time of the last activity in all-normal schedule.

Upon completion of the first round of mode elimination, the schedules containing the all-crashed and the all-normal modes are updated based on the remaining shortest and longest modes. Further elimination of long and short modes is carried out in accordance with the conditions outlined in Eqs. (8)-(14). This process of adaptive mode elimination is repeated until no further long or short modes can be eliminated using the aforementioned rules. As per the proposed optimization method, the simplified problem is utilized to identify the optimal solution for the current ε level. Following this, the value of ε is incremented by one, and the size reduction rules mentioned earlier are applied once again to reduce the search space. This process of mode elimination and optimization is then repeated iteratively until all feasible completion durations have been explored. Pseudo-code given in Fig. 1 describes the main steps involved in implementation of the proposed algorithm. The overview of this methodology is presented as a flowchart in Fig. 2. Based on the MILP formulation described in Eqs. (1)-(7) and the adaptive mode elimination rules presented in Eqs. (8)-(14), a Gurobi Optimization model is created using C# on Visual Studio 2019.

Algorithm (f)

```

1   $PF \leftarrow \emptyset$ 
2  While (  $\varepsilon \leq \varepsilon_{\max}$  ) {
3      Apply Mode Elimination rules (8)-(14) for  $\varepsilon$ 
4      Solve MILP formulation (1)-(7) for  $\varepsilon$ 
5       $Sol \leftarrow (f^{(1)}, f^{(2)})$ 
6      if ( $Sol \neq \text{nil} \wedge (\nexists Sol' \in PF : f_{Sol'}^{(2)} < f_{Sol}^{(2)})$ ) {
7           $PF \leftarrow (PF \cup \{Sol\}) \setminus \{Sol' \in PF : Sol \succ Sol'\}$ 
8           $\varepsilon \leftarrow \varepsilon + 1$ 
9      }
10 return ( $PF$ )
    
```

Fig. 1. Pseudo-code of the proposed optimization method.

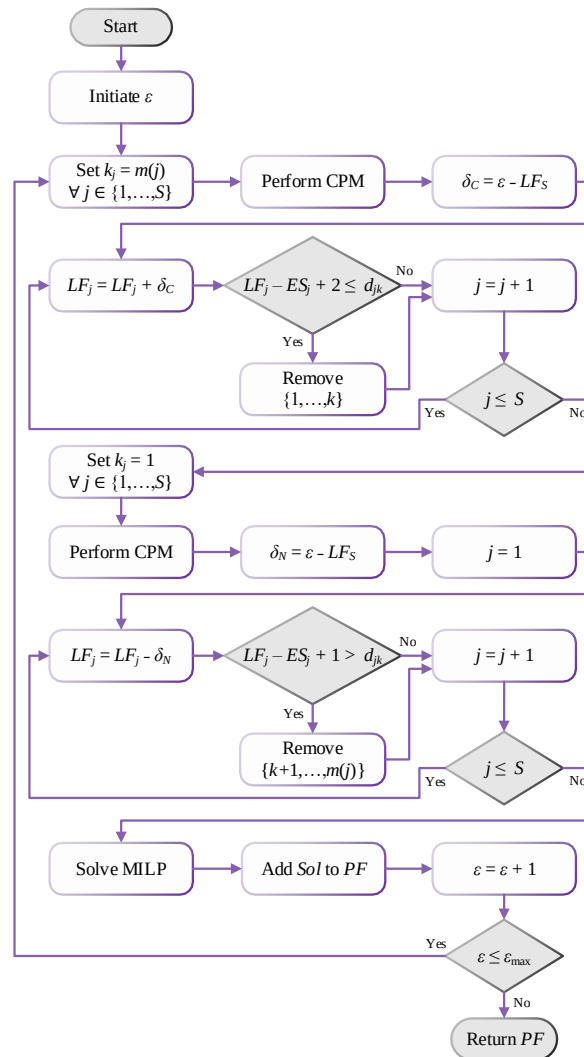


Fig. 2. Flowchart of the proposed optimization method.

3. Computational experiments

The proposed reduction strategy is experimented for optimization of a well-studied benchmark TCTP which was introduced by Hegazy [16]. The activity-on-node (AoN) representation as well as the time-cost modes of this problem are given in Fig. 3. As seen in Fig. 3, this TCTP contains 18 activities and up to five time-cost execution modes, with a single activity having two modes, ten activities with three

modes, two activities with four modes, and five activities having five modes. Therefore, this problem can be configured in numerous distinct mode combinations, resulting in 5,904,900,000 possible schedules. The effects of the adaptive mode elimination strategy on the performance of the proposed optimization method is explored by comparing the results over this benchmark TCTP by assuming an indirect cost of \$0/day.

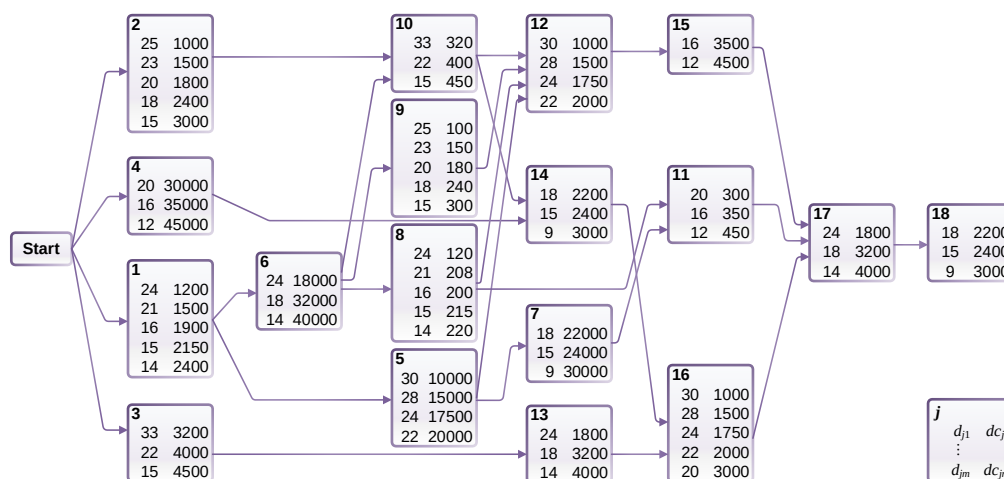


Fig. 3. Unreduced problem.

Table 1 provides the details of the obtained results by shedding light on the size of the reduced search spaces associated with every value of ϵ . In addition, a general overview of the impact of the mode elimination technique is covered in Table 2. As can be followed from Table 1, the ϵ -constraint-based method can achieve 44 non-dominated solutions ranging from (100 day, \$133,320) to (169 day, \$99,740). With reference to Table 2, size of the search space can be reduced at least by 96.29% down to size 218,700,000, for ϵ 's 118, 119, 120, 121, 122, 124, and 125. On the other hand, the size of the search space is reduced by a maximum amount of 99.99% down to size 48, at $\epsilon=100$. Figs. 4 and 5 are presented to exemplify how adaptive mode elimination removes the redundant modes at the two extreme points of $\epsilon=100$ and $\epsilon=169$, respectively. It should be pointed out that in these illustrations, the dimmed text indicates that the time-cost mode has been discarded. As can be followed from Fig. 4, there can be $1^{13} \times 2^4 \times 3^1$, i.e., 48 possible configurations when all combinations of the retained modes are considered. Similarly, it can be seen from Fig. 5 that for the reduced problem, a total of $1^{11} \times 2^7$, i.e., 128 different realizations are possible at $\epsilon=169$. Such remarkable drops in the search space complexity can potentially lead to significant reductions in computational burden of Pareto-oriented TCTP analysis.

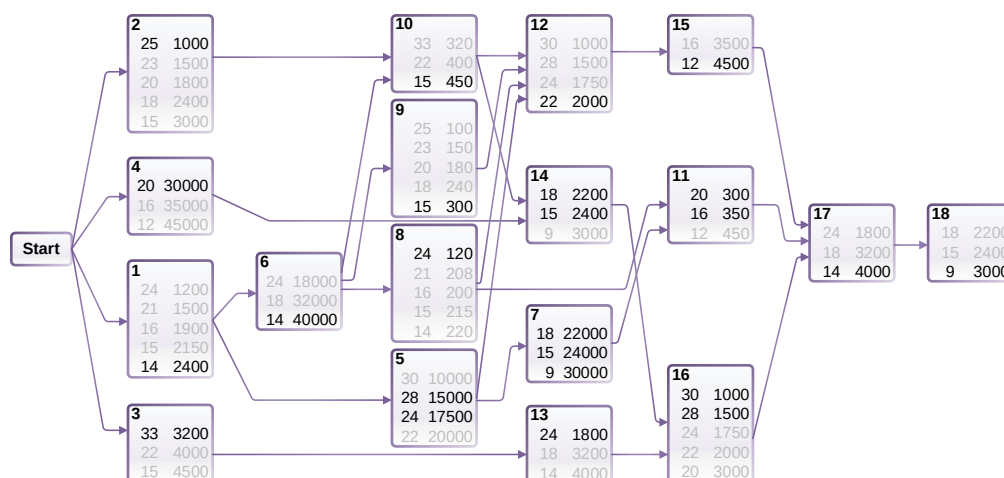


Fig. 4. Reduced problem at $\epsilon=100$.

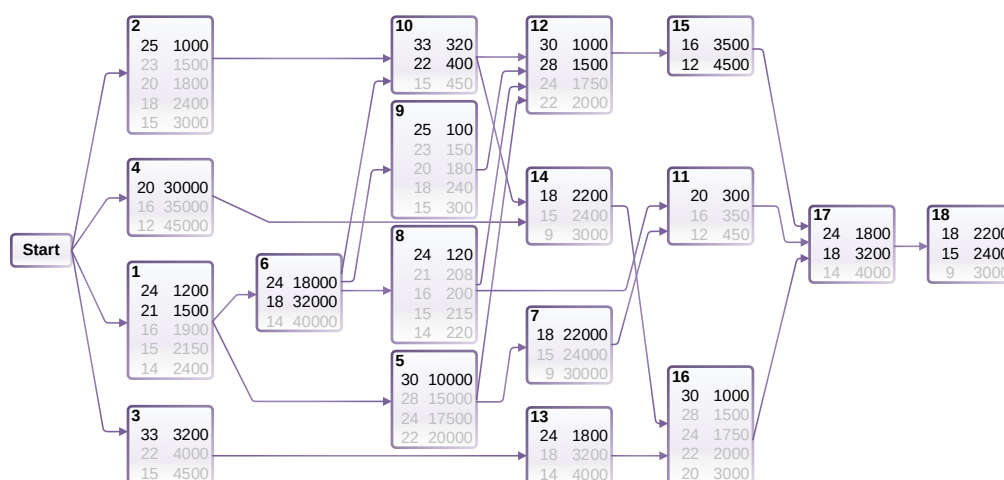


Fig. 5. Reduced problem at $\epsilon=169$.

The potential of the adaptive mode elimination is demonstrated by successfully reducing the search space of the employed problem and the results are summarized in Table 1. The first row of this table (shown in *italics*) presents the unreduced number of available activity modes and the corresponding size of search space for any duration between 100 and 169 days. The remainder of the table provides insight as to how the mode elimination mechanism enables reducing the search space per each level of ϵ . The computational experiment indicate that the proposed method is capable of locating Pareto front for this instance by searching merely a significantly small fraction of the search space. In fact, as summarized in Table 2, only 3,013,047,572 possible schedules are explored instead of 259,815,600,000, subsequent to reducing the problem per the proposed strategy. This is due to the fact that a conventional ϵ -constraint method would have explored $44 \times 5,904,900,000 = 259,815,600,000$ number of schedules for this problem had it not been reduced by the adaptive mode elimination strategy. It can therefore be claimed that the proposed method is capable of capturing the entire efficient frontier by searching only 1.16% of the unreduced search space required to be explored for the 44 Pareto solutions.

Table 1. Pareto front achieved through iterative problem size reduction.

Duration (day)	Total Cost (\$)	No. of Available Modes																		Search Space	Reduction (%)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
<i>100-169</i>	<i>133,320-99,740</i>	<i>5</i>	<i>5</i>	<i>3</i>	<i>3</i>	<i>4</i>	<i>3</i>	<i>3</i>	<i>5</i>	<i>5</i>	<i>3</i>	<i>3</i>	<i>4</i>	<i>3</i>	<i>3</i>	<i>2</i>	<i>5</i>	<i>3</i>	<i>3</i>	<i>5,904,900,000</i>	<i>0.00</i>
100	133,320	1	1	1	1	2	1	3	1	1	1	2	1	1	2	1	2	1	1	48	99.99
101	128,320	2	1	1	1	3	1	3	1	1	1	3	1	1	2	1	2	1	1	216	99.99
102	128,070	3	1	1	1	3	1	3	1	1	1	3	2	1	2	1	2	1	1	648	99.99
103	127,820	3	1	1	1	3	1	3	1	2	1	3	2	1	1	1	1	1	1	324	99.99
104	120,320	3	3	1	1	4	2	3	2	2	1	3	2	1	2	2	3	2	1	124,416	99.99
105	120,070	3	3	1	1	4	2	3	2	3	1	3	2	1	2	2	2	2	1	124,416	99.99
106	119,820	3	5	1	1	4	2	3	3	3	1	3	3	1	3	2	4	2	2	2,799,360	99.95
107	119,770	4	5	1	1	4	2	3	5	3	2	3	3	1	3	2	5	2	2	15,552,000	99.73
108	119,270	4	5	1	1	4	2	3	4	4	2	3	4	1	3	2	5	2	2	22,118,400	99.62
109	119,020	4	5	1	1	4	2	3	5	4	2	3	4	1	3	2	5	2	3	41,472,000	99.29
110	106,270	5	5	1	1	4	3	3	5	5	2	3	4	1	3	2	5	3	3	145,800,000	97.53
111	106,020	5	5	1	1	4	3	3	5	5	2	3	4	1	3	2	5	3	3	145,800,000	97.53
112	105,770	5	5	1	1	4	3	3	5	5	2	3	4	1	3	2	5	3	3	145,800,000	97.53
114	105,270	5	5	1	1	4	3	3	5	5	2	3	4	1	3	2	5	3	3	145,800,000	97.53
115	105,020	5	5	1	1	4	3	3	5	5	2	3	4	1	3	2	5	3	3	145,800,000	97.53
116	104,770	5	5	1	1	4	3	3	5	5	2	3	4	1	3	2	5	3	3	145,800,000	97.53
118	104,470	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29
119	104,220	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29
120	103,970	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29

Duration (day)	Total Cost (\$)	No. of Available Modes																		Search Space	Reduction (%)
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18		
121	103,820	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29
122	103,570	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29
124	103,070	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29
125	102,820	5	5	1	1	4	3	3	5	5	3	3	4	1	3	2	5	3	3	218,700,000	96.29
126	102,570	5	5	1	1	4	3	3	4	5	3	3	4	1	3	2	5	3	3	174,960,000	97.03
128	102,320	5	5	1	1	4	3	3	3	5	3	3	4	1	3	2	5	3	3	131,220,000	97.77
131	102,170	5	5	1	1	4	3	3	3	5	3	2	4	1	3	2	5	3	3	87,480,000	98.51
132	101,970	5	5	1	1	4	3	2	2	5	3	2	4	1	3	2	5	3	3	38,880,000	99.34
133	101,820	5	5	1	1	4	3	2	2	5	3	2	4	1	3	2	5	3	3	38,880,000	99.34
134	101,570	5	5	1	1	4	3	2	2	5	3	2	4	1	3	2	5	3	3	38,880,000	99.34
137	101,510	5	5	1	1	3	3	1	1	5	3	1	4	1	3	2	5	3	3	3,645,000	99.93
138	101,470	5	5	1	1	3	3	1	1	5	3	1	4	1	3	2	5	3	3	3,645,000	99.93
139	101,170	5	5	1	1	3	3	1	1	5	3	1	4	1	3	2	5	3	3	3,645,000	99.93
140	100,970	5	4	1	1	3	3	1	1	5	3	1	4	1	3	2	4	3	3	2,332,800	99.96
142	100,870	5	3	1	1	2	3	1	1	5	3	1	4	1	3	2	3	3	3	874,800	99.98
143	100,770	5	3	1	1	1	3	1	1	5	3	1	4	1	3	2	3	3	3	437,400	99.99
145	100,570	5	2	1	1	1	3	1	1	5	3	1	4	1	2	2	3	3	3	194,400	99.99
148	100,270	5	1	1	1	1	3	1	1	5	3	1	4	1	1	2	1	3	3	16,200	99.99
151	100,070	5	1	1	1	1	3	1	1	5	3	1	4	1	1	2	1	3	3	16,200	99.99
154	100,010	5	1	1	1	1	3	1	1	5	3	1	4	1	1	2	1	3	3	16,200	99.99
156	99,950	5	1	1	1	1	3	1	1	4	3	1	4	1	1	2	1	3	3	12,960	99.99
158	99,900	5	1	1	1	1	3	1	1	3	3	1	4	1	1	2	1	3	3	9,720	99.99
159	99,870	5	1	1	1	1	3	1	1	3	2	1	4	1	1	2	1	3	3	6,480	99.99
161	99,820	4	1	1	1	1	3	1	1	2	2	1	4	1	1	2	1	3	3	3,456	99.99
169	99,740	2	1	1	1	1	2	1	1	1	2	1	2	1	1	2	1	2	2	128	99.99

Table 2. Search space comparison for unreduced and reduced problems.

Description	Value
Size of smallest reduced search space	48
Size of largest reduced search space	218,700,000
Size of avg. reduced search space	68,478,354
Min reduction in search space (%)	96.29
Max reduction in search space (%)	99.99
Overall size of unreduced search space for locating 44 Pareto solutions	259,815,600,000
Overall size of reduced search space for locating 44 Pareto solutions	3,013,047,572
Percent reduction in overall size of search space for locating 44 Pareto Sols (%)	98.84

4. Conclusions

The significance of achieving an adequate balance between cost and duration of a project has been already well debated in the time-cost trade-off problem (TCTP) literature. However, studies focusing on this matter have scarcely addressed the potential of reduction techniques for Pareto front TCTPs. Of the problem size reduction technique, mode elimination has been the subject of research studies focusing on deadline and budget TCTPs but has never been deployed for solution of the more complex Pareto front variant. Accordingly, this study proposes a clever adaptive mode elimination strategy particularly designed for unravelling Pareto front TCTPs. The optimization model of the present study is based on the conventional ϵ -constraint method featuring the proposed clever adaptive mode elimination strategy. This optimization model is formulated and developed using mixed integer linear programming (MILP) and its effectiveness is experimented using a benchmark TCTP taken from the literature. It is shown that by leveraging this approach, the entire efficient frontier of the benchmark instance can be captured by exploring less than two percent of the overall unreduced search space. Such remarkable reductions in the solution space can potentially speed up the search and lower computational complexity

of Pareto front TCTPs that are inherently complicated and require decision-makers to consider multiple objectives instantaneously.

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